OpenSpec Plan: Delete Machine Exit Status 1

Change: **delete-running-vm-machine-exit-status-1** | Repo: **github.com/mkdev-ai/devsy-proxmox-provider**

Trigger: **error failed to delete machine: machineId=python-af390, err=run provider command "${PROXMOX_PROVIDER} delete" (): exit status 1**

Problem

Running devsy up (or an explicit provider delete) on a **running** workspace VM ends with a non-zero provider exit, surfaced to the user as exit status 1.

Live logs from the failing host confirm the root cause:

```
Error: deleting VM: API error: HTTP 501 — {"message":"Method 'POST /nodes/sojki/qemu/3006/destroy' not implemented","data":null}
```

Root cause (verified from live logs + main on default branch)

- **DeleteVM uses the wrong HTTP method.** pkg/proxmox/client.go POSTs to /nodes/{node}/qemu/{vmid}/destroy. Proxmox implements the VM-destroy API as **DELETE /nodes/{node}/qemu/{vmid}** with purge=1 and destroy-unreferenced-disks=1 query params (matching qm destroy --purge). The live host returns HTTP 501 "Method POST ... not implemented", so delete always exits 1 on a real Proxmox node.
- cmd/lifecycle.go also sleeps 2s and ignores StopVM, so it never waits for the shutdown task to finish. This is secondary (the 501 fires before the destroy path reaches a running-VM check), but it can cause transient 500s on the destroy when the VM is still up.

Goals

- devsy provider delete (and devsy up cleanup) exits 0 when the machine and its VM are removed on a real Proxmox VE host.
- DeleteVM calls the correct Proxmox API: DELETE /nodes/{node}/qemu/{vmid}, with purge=1 and destroy-unreferenced-disks=1 so the VM and disks are removed.
- The shutdown task is actually waited for before destroy (no transient 500s).
- A temporary retry is added for transient errors on destroy so a transient Proxmox 500 still lets delete succeed.

Non-goals

- Changing how Devsy names the provider binary or hosts it.
- Multi-cluster / cross-node deletion.
- Handling a VM stuck in a crashed state forever (timed out, not fixed).

Design summary

- **pkg/proxmox/client.go DeleteVM:** DELETE /nodes/{node}/qemu/{vmid} with purge=1 + destroy-unreferenced-disks=1. Add transient-error retry (1s/2s/4s, max 3) and surface the Proxmox API error detail in the wrapped error.
- **cmd/lifecycle.go runDelete:** replace the sleep + ignored StopVM with a real WaitForTask on the shutdown UPID. Only call DeleteVM(..., true) after the shutdown task reports stopped/OK. Keep the emergency-cleanup step (stop, destroy without purge).
- **Tests:** contract cases for running-VM delete, transient 500 retry, and an end-to-end TestRunDelete.

Implementation tasks

- [ ] Write a failing unit test that mimics a running VM (destroy returns HTTP 501/500) and confirm the current code returns exit status 1.
- [ ] Fix the destroy endpoint: DELETE /nodes/{node}/qemu/{vmid} with purge=1 + destroy-unreferenced-disks=1 (replace the POST /destroy path).
- [ ] Change the HTTP method to DELETE and add a transient-error retry (1s/2s/4s, max 3) for HTTP 5xx / API errors on the destroy request.
- [ ] Surface the underlying Proxmox API error detail in the wrapped error.
- [ ] Remove the time.Sleep(2 * time.Second) hack and the ignored StopVM call in runDelete.
- [ ] Wait for the shutdown task with WaitForTask (capped at ~30s) and only proceed to destroy if the task reports stopped + OK.
- [ ] Keep the emergency-cleanup step (stop, destroy without purge) after a transient destroy failure.
- [ ] Add contract tests: TestDeleteVM_RunningVM, TestDeleteVM_Transient500, TestRunDelete.
- [ ] Extend env-gated acceptance (TestDevsyAcceptance): create a devsy-named VM, start it, delete it, assert no leftover VM and no orphaned finalizer.
- [ ] Run go test -tags=acceptance -v -run TestDevsyAcceptance with DEVSY_PROXMOX_* set and verify exit 0.
- [ ] Update README.md delete/cli docs to describe running-machine delete and emergency cleanup.

Plan produced from verified reads of live logs and main. Artifacts: proposal.md, design.md, tasks.md, spec.md, plan.pdf, plan.html. Go implementation + tests are pending (not yet written).